PHISHING DETECTION AND AWARENESS REPORT

Prepared by: Edwin Baiju
Cyber Security Internship, Future Interns (Task 2)
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EXECUTIVE SUMMARY

Phishing is a type of cyber attack where criminals send fake emails that look genuine, to trick people into sharing passwords, money or personal details. It is one of the most common ways businesses get hacked, and a single click by one employee can expose a whole company.

In this report, 7 emails were analyzed for warning signs. Of these, 3 were classified as Phishing, 2 as Suspicious and 2 as Safe. The report explains in simple language how each email was judged and ends with practical steps that staff can follow to stay protected.

METHODOLOGY

1. Six simulated emails were created for training, covering common phishing styles (fake bank or payment alerts, fake IT messages, parcel scams, fake invoices and fake job offers). One real email was also checked using its authentication results.
2. Each email was checked for these indicators: sender address, lookalike domains, urgent or threatening language, mismatched links, generic greetings, unexpected attachments and requests for sensitive data.
3. For the real email, the header was opened in Gmail ("Show original") to check SPF, DKIM and DMARC results.
4. Each email was classified as Safe, Suspicious or Phishing. Phishing means clear signs of fraud. Suspicious means warning signs that need confirmation. Safe means no warning signs and passed authentication.
5. All links were defanged ([.] and hxxp) so no one can click them by accident. No links or attachments from phishing emails were opened.

SECTION 1: SAMPLE EMAIL ANALYSIS

(Note: Emails 1-6 are simulated samples created for educational purposes. Links are defanged, with [.] replacing dots, as is standard security practice.)

EMAIL 1
From: PayPal Support <support@paypa1-secure[.]com>
Subject: Account limited! Verify within 24 hours
Body: Dear Customer, we noticed unusual activity. Your account will be closed in 24 hours unless you verify.
Link shown: www.paypal.com/verify
Real destination: hxxp://paypa1-secure[.]com/login
Verdict: PHISHING

Reason: Lookalike domain (the number 1 replaces the letter l), urgent threat, link text does not match the real destination, generic greeting.

EMAIL 2
From: IT Helpdesk <it-support@company-helpdesk[.]xyz>
Subject: Your password expires today
Body: Click below to keep your access. Failure to act will lock your account.
Real destination: hxxp://company-helpdesk[.]xyz/reset
Verdict: PHISHING
Reason: Unknown .xyz domain, urgency, asks for login details, not a real internal address.

EMAIL 3
From: Express Delivery <track@parcel-info[.]top>
Subject: Parcel on hold, pay Rs.49 fee
Body: Your parcel could not be delivered. Pay a small fee to reschedule.
Link: hxxp://bit[.]ly/3xAmpl3
Verdict: PHISHING
Reason: Shortened link hides the destination, small payment request, unknown domain.

EMAIL 4
From: Accounts, ABC Traders <abctraders.accounts@gmail[.]com>
Subject: Invoice attached, please pay today
Body: Please find the invoice and pay today to avoid delay.
Attachment: invoice.zip
Verdict: SUSPICIOUS
Reason: Business using a free email address, ZIP attachment, pressure to pay quickly. Needs confirmation with the vendor.

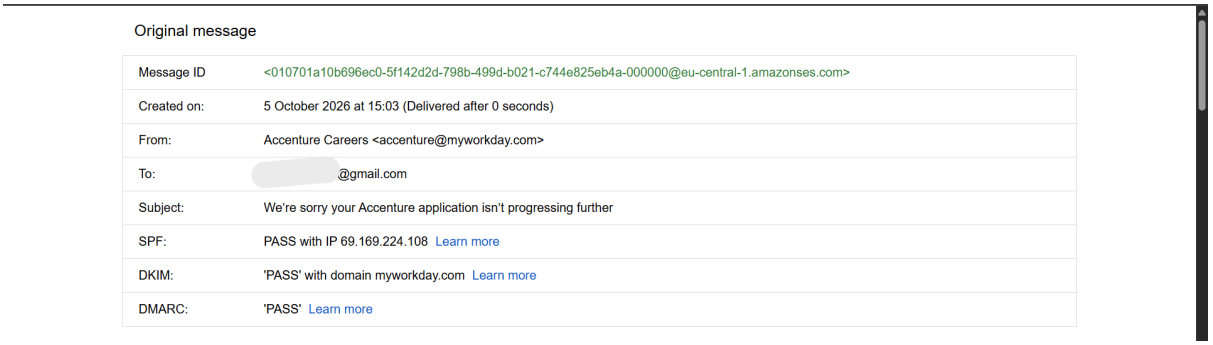
EMAIL 5
From: HR Recruiter <jobs.hr2026@outlook[.]com>
Subject: You're selected!
Body: Congratulations! Reply with your ID proof and bank details to get your offer letter.
Verdict: SUSPICIOUS
Reason: Generic free email address, asks for sensitive data, no official company domain.

EMAIL 6
From: GitHub <noreply@github[.]com>
Subject: A new sign-in to your account
Body: Hi (your username), we noticed a new sign-in from Chrome on Windows. If this was you, no action is needed.
Link: https://github.com/settings/security
Verdict: SAFE
Reason: Correct official domain, personalized greeting, no urgency, no unusual links.

EMAIL 7 (REAL EMAIL, AUTHENTICATION CHECK)
From: Accenture Careers <accenture@myworkday[.]com>
Subject: We're sorry your Accenture application isn't progressing further

Verdict: SAFE
Reason: The sender domain differs from the company name, but it is the hiring platform Workday, which companies use to send emails on their behalf. The email passed SPF, DKIM and DMARC authentication, there are no links asking for personal data, no urgency, and no requests for sensitive information. Lesson: a different domain alone does not mean phishing, so always check the authentication results as well.

Screenshot: Email header authentication results (SPF, DKIM, DMARC all PASS)



Original message

Message ID	<010701a10b696ec0-5f142d2d-798b-499d-b021-c744e825eb4a-000000@eu-central-1.amazonses.com>
Created on:	5 October 2026 at 15:03 (Delivered after 0 seconds)
From:	Accenture Careers <accenture@myworkday.com>
To:	@gmail.com
Subject:	We're sorry your Accenture application isn't progressing further
SPF:	PASS with IP 69.169.224.108 Learn more
DKIM:	'PASS' with domain myworkday.com Learn more
DMARC:	'PASS' Learn more

COMMON PHISHING TECHNIQUES EXPLAINED SIMPLY

1. Spoofed sender: The email looks like it comes from a trusted name (a bank, a delivery company, your boss), but the real address is different. Always check the actual email address, not just the display name.

2. Lookalike domains: Attackers register websites that look almost identical to real ones, such as paypa1.com instead of paypal.com. At a quick glance, the difference is easy to miss.

3. Urgency and fear: Messages like "Your account will be closed in 24 hours" push people to act without thinking. Real companies rarely threaten you by email.

4. Malicious links: The link text shows one website, but clicking takes you to another. Hovering over a link (without clicking) reveals the true destination.

5. Dangerous attachments: Files such as .zip, .exe or fake invoices can install malware on your computer when opened.

6. Requests for sensitive data: Genuine organizations do not ask for passwords, OTPs, ID proof or bank details over email.

BUSINESS RISK
A successful phishing attack can lead to stolen login details, financial loss, leaked customer data, malware infection, damage to company reputation, and legal or regulatory trouble.

Because phishing targets people rather than systems, even a company with strong technical security can be affected.

PREVENTION AND AWARENESS GUIDELINES FOR USERS

Before you click:
- Check the sender's full email address, not just the name.
- Hover over links to see where they really go.
- Be careful with urgent, threatening or "too good to be true" messages.
- Never open unexpected attachments, especially ZIP or EXE files.

Protecting your accounts:
- Never share passwords or OTPs by email, phone or message.
- Turn on two-factor authentication (2FA) on all important accounts.
- Use a different strong password for each account, with a password manager if possible.
- Type the website address yourself or use a saved bookmark instead of clicking email links.

If you are unsure:
- Do not reply, click or download anything.
- Contact the company through its official website or phone number.
- Report the email to your IT or security team.
- Use "Report phishing" in Gmail or Outlook, then delete the email.

If you already clicked:
- Disconnect from the network if you opened an attachment.
- Change your password immediately and tell your IT team.
- Watch your accounts for unusual activity.

CONCLUSION

Phishing works by exploiting trust and urgency, not by breaking technology. This report showed that most phishing emails can be spotted by checking the sender address, links, language and authentication results (SPF, DKIM, DMARC). Email 7 also showed that an unfamiliar domain is not always dangerous, so decisions should rely on several signs together. With regular awareness training and the simple habits listed above, organizations can greatly reduce the risk of a successful phishing attack.